

COEN 390/ELEC 390 - Computer and Electrical Engineering Product Design Project ©

Project Document Summer 2026

For your project your team must build an Android app. Your app must use an external sensor. You must identify and interview stakeholders for the product you are building to elicit ideas for your project.

Teams:

This is a team project. Teams should have five members and at least one person with reasonably good programming experience. Teams should be comprised of COEN and ELEC students. We may have a small number of teams with six members. There will be no teams with 4 or 5 members. Almost all teams will be 5 members.

Students may form their own teams. If you are forming your own team you should sent me (via email) the team members by Friday May 22, 2026. Indicate in that email which student (s) have reasonably good programming experience. Also indicate what program each prospective team member is in. I will have to approve all teams.

If you would rather that I place you on a team (perhaps to meet new people in your class), you need do nothing. Everyone who has not been part of a submitted team will be put on teams after the deadline.

Blog:

Each team will keep a Blog. The Blog should be constructed using the Excel sheet that will be provided. This Bog will continue to grow throughout the term. It must be submitted with each milestone of the project. The final version of the Blog will need to be endorsed by each team member. For this each team member should create a page where they say, "I have reviewed the team blog and I agree it is accurate of what we did." And then sign your name.

Potential Product Ideas:

Really good products will be ones that are original and where you have benefited from and used input from stakeholders.

The University has a lot of individuals working in many interesting areas which could benefit from well executed apps. I'd encourage you to spend time in the first few weeks of the term, prior to the submission of Milestone 1 identifying potential products and stakeholders from various parts of the University.

Interesting areas include:

1. Health and fitness. The PERFORM center on the Loyola campus has a focus on healthy aging. Its research efforts span a range of disciplines including engineering.
2. Smart Homes and Smart Cities. Smart here refers to sensors in cities or homes that are used to provide services to people. As you go through your daily life think about some things that frustrate you or that you wish could be better. In the past we've had projects around cooking meat to a desired doneness, and another to help individuals who do their laundry in the laundry room of a large apartment building.
3. Concordia Applied AI Institute: This institute has many projects that involve around the implication of AI to solve real world problems.

The very best projects of the past have been ones that are original. Thus, just don't take an example from above, but find for yourself a new problem to solve. In that vein we've had a large number of plant monitoring projects many of which are not so inspiring. I'd suggest that unless you have something truly amazing and original that you plan to do for this topic, that you avoid a plant monitoring project.

Meetings:

During the sprints, there are 4 formal meetings that each team must attend with all members present.

- a) At the beginning of Sprint 1
- b) At the end of Sprint 1 / beginning of Sprint 2
- c) At the end of Sprint 2 / beginning of Sprint 3
- d) At the end of Sprint 3. This is your final demo and presentation.

Meetings a) b) and c) will **be done remotely unless otherwise specified by the Scrum Master**. Meeting d), the final demo and presentation, **is done in person**.

In meetings b) c) and d) each team will do a demo of what they have built. You should plan to have one of your team members have all of the hardware components so that they can demonstrate the hardware.

Scrum Masters:

Each team will have a Scrum Master (please see book, notes and lectures for the role of Scrum Master). Each group will be assigned their particular Scrum Master when the Milestone 1 submissions are returned.

Document Submission:

You will submit all documents on Moodle. Aside from code, all other documents for each milestone/sprint should be submitted in one comprehensive pdf file. **You must submit actual documents. Links to documents are not accepted.**

Milestone 1 – Planning - Due May 29, 2026

The team should submit

- a. At least 6 opportunity statements (see Lesson 1 and Appendix A of this document)
- b. An evaluation of the opportunity statements
- c. A ranking of the opportunity statements
- d. A mission statement of the top ranked opportunity statement (see Lesson 1 and Appendix B of this document)
- e. The Team Blog
- f. An expectation of originality form

Milestone 2 – Requirements Doc - June 12 , 2026

The team should submit

- a. Information on at least 2 stakeholders that were interviewed. Must have at least 1 interview for each stakeholder. Include date and time you met each. To maintain privacy, give a fake name, or first name only.
- b. Your prepared a consent form for those you interviewed (just the form, not filled in by your interviewees).
- c. Interview script prepared (either one global or one for each stakeholder tailored to that stakeholder).
- d. Give the 3 most important outcomes each team member got from each interview. (Note you do **not** need to provide the detailed answers provided by the interviewees, only your team members' most important takeaways from each interview.)
- e. Product Backlog. Stories may still be at a high level of abstraction (epics). Hopefully at least some have begun to be broken down into smaller stories.
- f. A one-page submission on the ethical dimensions of your product (potential for misuse, privacy issues, etc.).
- g. The Team Blog.
- h. An expectation of originality form.

Sprints:

As a rule of thumb, you should aim to have 20 hours of work per team mate in each two-week sprint. Not all of this will be spent developing code (some time will be spent preparing the product backlog for the next meeting), but a very large proportion should.

One of your teammates should be designated the product owner. This person must have a vision of what the product is that you are trying to build. Generally, she/he does not decide how to build things, but what to build. He/she has core responsibility for organizing meetings to groom the product backlog, sets the initial sprint goals, provides clarifying details, and finally does acceptance tests on code once it is “done”. This person should also have some coding to do themselves although it can be modest.

You should also plan to do a “Scrum” 3 times in each sprint. Recall these scrums which are done daily when one is working every day, are short (ideally 15 minutes) meetings where all are present. These are generally face to face meetings. Given that you do not have a central office where you go to work each day you may wish to do these meetings using a tool where you all communicate at the same time, but perhaps from different locations. Each member briefly brings other team mates up to date on: a) what they have done since the last scrum, b) what they will do next, c) outline any impediments that are preventing or delaying their work. Team members pass on comments (if they are short) to each other. Longer comments should be taken offline so as not to take up everyone’s time.

Sprint 1 – June 19 - July10, 2026

Meeting with the Scrum Master at beginning of sprint. The meeting will be arranged with the Scrum Master for each team. Title of Meeting Sprint 1 Planning.

Before this meeting you should submit:

- Product backlog. By this time the team should have started to break down some of the items in the product backlog that are going to be worked on in Sprint 1.
- Preliminary Sprint 1 goal.
- Preliminary Sprint 1 backlog.
- Sketch of the screens your app will have along with indications of the navigation (how one will go from one screen to another).

At this meeting with the Scrum Master you will

- Finalize the Sprint 1 goal
- Finalize the Sprint 1 backlog.

Sprint 2 – July 13 - July 24, 2026

Meeting with the Scrum Master at the beginning of Sprint 2. Title of Meeting Wrap up for Sprint 1 and planning for Sprint 2

Before this meeting you should submit:

- Design Document (a template will be provided).
- Definition of Done (See chapter 4 of the textbook and slides).
- Test document including results for Sprint 1.
- Product backlog. Include items completed in a separate list. Some items should be broken down in preparation for Sprint 2.
- Preliminary Sprint 2 goal.
- Preliminary Sprint 2 backlog.

At this meeting with the Scrum Master you will

- Demonstrate your app.
- Discuss the product and process with the scrum master
- Finalize the Sprint 2 goal.
- Finalize the Sprint 2 backlog.

Sprint 3 – July 27 - August 7, 2026

Meeting with Scrum Master at the beginning of Sprint 3. Title of Meeting Wrap up for Sprint 2 and planning for Sprint 3

Before this meeting you should submit:

- Updated Design Document.
- Definition of Done.
- Summary of testing including results for Sprint 2.
- Product backlog. Include items completed in a separate list. Some items should be broken down in preparation for Sprint 3.
- Preliminary Sprint 3 goal.
- Preliminary Sprint 3 backlog.

At this meeting with the Scrum Master you will

- Demonstrate your app.
- Discuss the product and process with the Scrum Master
- Finalize the Sprint 3 goal.
- Finalize the sprint 3 backlog.

Final Oral Presentation and Demo: Week of August 10

In this group presentation you should

- a. Identify your customer.
- b. Explain the value of your product for the customer
- c. Present how your product function. This might include the architecture of your product including where important processing occurs (phone, controller board, server) and the communication links. It should also contain information on important signal processing and algorithms in your product.

The presentation should be between 10 and 12 minutes long. In addition, at this meeting, you will give a final demo. The final demo and presentation will be done in person. Note that the phones should be turned in at the end of the demo.

Final submission of Project Due August 14, 2026

The team should submit

- a. Mission statement (updated if necessary). Note list your initial assumptions. By this time, they should have been either validated as true or found to be false. Be clear for each assumption, if they were true or false. If an assumption was false state how that changed your product development.
- b. Final product backlog. Include completed items in a separate list.
- c. Design document (revised)
- d. User manual (if external documentation is necessary)
- e. Definition of Done (updated if necessary)
- f. Test document
- g. A two-page report on the ethical dimensions of your product and what you did or could do to mitigate these.
- h. A 2-min video where you “pitch” your product. That is where you try to convince potential investors/partners to invest in your product. Be sure you explain what your product does, who your target audience is, and what benefits your product will give to your customers and how your product will be profitable. **The Department may make your video available on a public 390 website, which future 390 students and others can look at.**
- i. The slides (converted to pdf) from your final oral presentation.
- j. Your product App.
- k. The final Team Blog.
- l. An expectation of originality form.

Coding Help Available

Once the sprints start, each group will be entitled to two 1.5 hour meetings with one of the TAs to receive technical help on creating your app. One of the TAs will contact the group when this service is ready.

Appendix A. Opportunity statement.

An Opportunity Statement is a short coherent statement describing the opportunity.

Example of an opportunity statement.

“Develop a new black and white (B&W), digital, networkable, document center platform for the office market, including scanning, storage, fax, distribution and printing capabilities.”

For your project, the opportunity statement should clearly state the customers you are aiming at.

Appendix B. Mission Statement

Provide clear guidance for the product development team by formulating a detailed definition of the target market and of the assumptions under which the development team will operate. These decisions are captured in a **Mission Statement**

Mission Statement

Product Description: Brief (1-sentence) The “stress prompter app” will measure heart rate and potentially other body signals and prompt the user to do yoga or other stress reducing exercises when appropriate.

Benefit Proposition: Stress has many negative health consequences, reducing it promptly will yield a happier and healthier life. Also, when stressed people often make bad decisions, reminding someone to reduce their stress may prevent them from making these bad decisions. Other stress apps require the user to do something to have their stress tested. In contrast our app constantly monitors the user’s stress prompting them when it is too high.

Key business goals: The app will require body sensors totalling \$10 (including distribution costs). We plan to sell this app for \$15 yielding a \$5 per unit sold gross margin (profit per unit less costs per unit). Our team will spend 300 hours creating this app at \$50 per hour or \$15000 cost. We will need to sell 3000 units to break even. Because stress is widespread and many are aware of its effects we believe we can exceed our break even target. There may also be potential for ads for stress relieving products or services that may be of interest to users.

Target market: Our initial target market will be students who periodically experience stress (exams projects etc.). Students are more likely to adapt new technological products than other users and can provide good word of mouth advertising to other potential users. They also will eventually graduate and then start a long career filled with stressful situations.

Assumptions: Assumption 1: we can find a comfortable, and if necessary stylish sensor that users will be willing to wear. Assumption 2: etc.

Constraints: Constraint 1: Project must be complete by April 11, 2024. Constraint 2: etc.

Stakeholders: 1) users who will pay for and use the project 2) distributors for the software and the sensors 3) suppliers for the sensors 4) the app store/google play store 5) providers of stress related products or services who may advertise on our app 6) Eventually medical professionals may use our app in medical settings to monitor patients.